University of California, Davis
Department of Agricultural and Resource Economics
ARE 252 – Optimization with Economic Applications – Lecture Notes 12
Quirino Paris

1. Spatial Equilibrium – Behavioral Hypotheses . . . . . page 1
2. Spatial Equilibrium – Perfect Competition 2
3. Spatial Cartel Equilibrium: Monopoly – Perfect Competition 3
4. Spatial Cartel Equilibrium: Monopoly – Monopsony 5
5 Spatial Cournot—Nash Equilibrium: Oligopoly–Perfect Competition 6
6. Spatial Cournot—Nash Equilibrium: Oligopoly–Oligopsony 9
7. Numerical Examples of Spatial Equilibrium 10

1. Spatial Equilibrium – Behavioral Hypotheses

Spatial equilibrium deals with a section of economics that attempts to explain the trade flow of commodities and their price formation among producing and consuming regions. In general, it involves three categories of economic agents: consumers, producers and traders. From a behavioral perspective, consumers are considered price takers who express their demand for a commodity by means of an aggregate demand function. Producers can be considered either price takers or agents who may behave according to imperfect competition rules. When producers are price takers, their final result is expressed as an aggregate supply function for a given commodity. Traders may behave as oligopsonists (monopsonists in the limit) on the regional supply markets and either as oligopolists or monopolists (cartels) on the regional demand markets. Often, producers and traders are considered as the same economic agents. The limit case of perfect competition among all regions implies that there are no identifiable traders: commodities are transferred from producing to consuming regions by the action of the "invisible hand."

Given the vast range of behaviors characterizing spatial equilibrium, we will limit the analysis to five combinations of behavioral rules: (a) perfect competition on both the supply and consumption markets; (b) perfect competition on the supply market and cartel behavior (monopoly) on the export/consumption markets; (c) cartel behavior (monopsony) on the supply market and cartel behavior (monopoly) on the export/consumption market; (d) perfect competition on the supply market and oligopoly (Cournot-Nash equilibrium) on the export/consumption markets; (e) oligopsony (Cournot-Nash equilibrium) on the supply market and oligopoly on the export/consumption market. The Cournot-Nash equilibrium refers to non-cooperative oligopoly and oligopsony firms: each Nash oligopoly (oligopsony) firm makes production and profit-maximizing decisions assuming that its choices do not affect oligopolists' (oligopsonists') decisions in other regions.

We consider the production and exchange of only one commodity among R regions. The extension to more than one commodity is straightforward. We assume knowledge of a linear inverse demand function for each region

$$p_j^D = a_j - D_j x_j^D \qquad j = 1,...,R \tag{1}$$

where p_j^D and x_j^D are price and quantity demanded in the j-th region. The known coefficients $a_j > 0$ and $D_j > 0$ are the intercept and slope of the demand function, respectively. We assume

knowledge also of a linear supply function for each region. This function can also be regarded as the marginal cost (MC_i) function for each region

$$p_i^S = b_i + S_i x_i^S = MC_i \qquad i = 1,...,R \tag{2}$$

where p_i^S and x_i^S are price and quantity supplied in the i-th region. The known coefficients b_i and $S_i > 0$ are the intercept and slope of the supply function, respectively. Bilateral unit transaction costs are also known for all pairs of regions and are stated as t_{ij} .

2. Perfect Competition Spatial Equilibrium

Following Samuelson (1952), Takayama and Judge (1964), and many other authors since, the primal specification of this spatial equilibrium is stated as

$$\max QWF = \sum_{j=1}^{R}(a_j - D_j x_j^D/2)x_j^D - \sum_{i=1}^{R}(b_i + S_i x_i^S/2)x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij}x_{ij} \tag{3}$$

subject to $D \le S$ dual variables

$$x_j^D \le \sum_{i=1}^{R} x_{ij} \qquad \text{regional demand} \qquad p_j^D \ge 0 \tag{4}$$

$$\sum_{j=1}^{R} x_{ij} \le x_i^S \qquad \text{regional supply} \qquad p_i^S \ge 0 \tag{5}$$

where all the variables are nonnegative. *QWF* stands for quasi-welfare function and measures the sum of the consumer and producer surpluses netted out of transaction costs. The first term of equation (3) represents the sum of the integrals under the demand functions of all regions while the second term is the sum of the integrals under the marginal cost functions (the inverse supply functions). The third term represents total transaction costs. The solution of model [(3)-(5)] produces optimal quantities demanded, supplied and traded among regions as well as equilibrium demand and supply prices as the Lagrange multipliers (dual variables) of constraints (4) and (5), respectively. In this model, profit of the atomized producing firm is equal to zero. The relevant KKT conditions are derived from the Lagrange function of problem [(3)-(5)]

$$\begin{aligned} L = &\sum_{j=1}^{R}(a_j - D_j x_j^D/2)x_j^D - \sum_{i=1}^{R}(b_i + S_i x_i^S/2)x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij}x_{ij} \\ &+ \sum_{j=1}^{R} p_j^D\left(\sum_{i=1}^{R} x_{ij} - x_j^D\right) + \sum_{j=1}^{R} p_i^D\left(x_i^S - \sum_{j=1}^{R} x_{ij}\right) \end{aligned} \tag{6}$$

with

$$\frac{\partial L}{\partial x_j^D} = a_j - D_j x_j^D - p_j^D \le 0 \qquad \text{dual constraints} \tag{7}$$

$$\frac{\partial L}{\partial x_i^S} = -b_i - S_i x_i^S + p_i^S \le 0 \qquad \text{dual constraints} \tag{8}$$

$$\frac{\partial L}{\partial x_{ij}} = p_j^D - p_i^S - t_{ij} \le 0 \qquad \text{dual constraints} \tag{9}$$

Each of these KKT conditions has the structure of $MR \leq MC$ (eliminating the negative signs). In particular, relation (9) takes on the form of

$$p_j^D \leq p_i^S + t_{ij}$$

$$(a_j - D_j x_j^D) \leq (b_i + S_i x_i^S) + t_{ij}$$

which establishes that, for an equilibrium (solution) of the problem, the destination price must always be less-than-or-equal to the origin price plus the unit transaction cost between region *i* and region *j*. Figure 1 illustrates this important economic relation.

3. Spatial Cartel Equilibrium: Monopoly – Perfect Competition

When exporters collude, a cartel is formed. The intent of a cartel is to maximize total aggregate profit for the cartel members. The behavior of cartel members, therefore, is to maximize the joint profit by selling the monopoly output in each region at the monopoly price. Hence, the spatial monopoly model assumes that, in all regions, output is controlled by one agent, that is, the cartel (no cheating is assumed or allowed). On the supply side we assume perfect competition behavior.

Algebraically, this cartel (monopoly-perfect competition) model varies only slightly – but very significantly in an economics sense – from the perfect competition model of section 2. For example, the monopoly-perfect competition objective function is stated as

$$\begin{aligned} \max Cartel\pi &= \sum_{j=1}^{R} p_j^D x_j^D - \sum_{i=1}^{R} (b_i + S_i x_i^S / 2) x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} \\ &= \sum_{j=1}^{R} (a_j - D_j x_j^D) x_j^D - \sum_{i=1}^{R} (b_i + S_i x_i^S / 2) x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} \end{aligned} \tag{10}$$

$Cartel\pi$ stands for cartel profit. The objective function (10) differs from the objective function (3) only by the coefficient (1/2) in the revenue part of profit (the first term). The primal version of the spatial cartel model is constituted by equation (10) and constraints (4) and (5). The derivation of the relevant KKT conditions reveals the structure of the marginal cost of the cartel-exporter. In this case, the Lagrange function is stated as

$$\begin{aligned} L = &\sum_{j=1}^{R} (a_j - D_j x_j^D) x_j^D - \sum_{i=1}^{R} (b_i + S_i x_i^S / 2) x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} \\ &+ \sum_{j=1}^{R} \rho_j \left(\sum_{i=1}^{R} x_{ij} - x_j^D \right) + \sum_{j=1}^{R} p_i^D \left(x_i^S - \sum_{j=1}^{R} x_{ij} \right) \end{aligned} \tag{11}$$

In this spatial cartel model, the Lagrange multipliers of the demand constraints (4) have been chosen as $\rho_j \neq p_j^D$ because the monopoly marginal revenue is different from the monopoly price. Then, the relevant KKT conditions take on the following structure:

$$\frac{\partial L}{\partial x_j^D} = a_j - 2D_j x_j^D - \rho_j \leq 0 \tag{12}$$

$$\frac{\partial L}{\partial x_i^S} = -b_i - S_i x_i^S + p_i^S \le 0 \tag{13}$$

$$\frac{\partial L}{\partial x_{ij}} = \rho_j - p_i^S - t_{ij} \le 0 \tag{14}$$

Assuming that each region will have a positive demand, $x_j^D > 0$, relation (12) will turn into an equation (by complementary slackness conditions) and, thus, $\rho_j = a_j - 2D_j x_j^D$ which, in turn, will induce relation (14) to take on the following structure

$$\begin{aligned} & a_j - 2D_j x_j^D - p_i^S - t_{ij} \le 0 \\ & a_j - D_j x_j^D - D_j x_j^D - p_i^S - t_{ij} \le 0 \\ & p_j^D - D_j x_j^D \le p_i^S + t_{ij} \\ & \quad MR \ \le \ MC \end{aligned} \tag{15}$$

This means that the monopoly price p_j^D (for the activated route connecting regions $i-j$) is equal to the marginal cost ($p_i^S + t_{ij}$) plus the segment $D_j x_j^D$ (market power) as indicated in figure 1, often called "monopoly power." The monopoly profit of the j-th region is given by the sum of areas *A* and *B*.

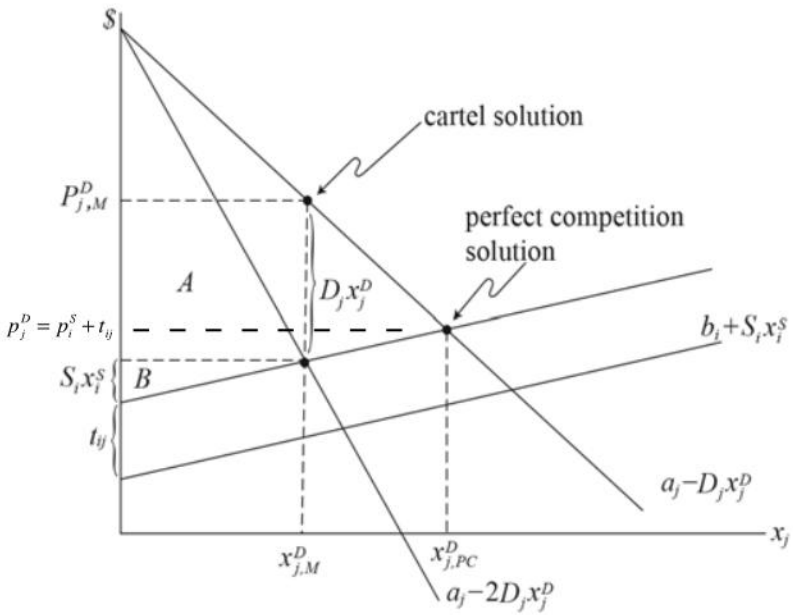


Figure 1. Cartel (*M*) and perfect competition (*PC*) solutions

4. Spatial Cartel Equilibrium: Monopoly – Monopsony

Firms can collude also on the production side. In this case, the cartel behavior assumes the form of a monopsonist. We assume that the cartel behaves as a monopsony on the supply market and as a monopolist on the export/consumption market. The objective function of this model is stated as

$$\begin{aligned}\max Cartel\pi &= \sum_{j=1}^{R} p_j^D x_j^D - \sum_{i=1}^{R} p_i^S x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} \\ &= \sum_{j=1}^{R} (a_j - D_j x_j^D) x_j^D - \sum_{i=1}^{R} (b_i + S_i x_i^S) x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij}\end{aligned} \tag{16}$$

subject to the usual demand and supply constraints (4) and (5). In this case, however, the Lagrange multipliers (dual variables) are different from demand and supply prices. Hence, the Lagrange function and the relevant KKT conditions take on the following structure:

$$\begin{aligned}L = &\sum_{j=1}^{R} (a_j - D_j x_j^D) x_j^D - \sum_{i=1}^{R} (b_i + S_i x_i^S) x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} \\ &+ \sum_{j=1}^{R} \rho_j \left(\sum_{i=1}^{R} x_{ij} - x_j^D \right) + \sum_{j=1}^{R} \phi_i \left(x_i^S - \sum_{j=1}^{R} x_{ij} \right)\end{aligned} \tag{17}$$

with relevant KKT conditions as

$$\frac{\partial L}{\partial x_j^D} = a_j - 2D_j x_j^D - \rho_j \le 0 \tag{18}$$

$$\frac{\partial L}{\partial x_i^S} = -b_i - 2S_i x_i^S + \phi_i \le 0 \tag{19}$$

$$\frac{\partial L}{\partial x_{ij}} = \rho_j - \phi_i - t_{ij} \le 0 \tag{20}$$

Assuming that each region will have a positive demand, $x_j^D > 0$, and a positive supply, $x_i^S > 0$, relations (18) and (19) will turn into equations (by complementary slackness conditions) and, thus, $\rho_j = a_j - 2D_j x_j^D$ and $\phi_i = b_i + 2S_i x_i^S$ which, in turn, will induce relation (20) to take on the following structure

$$\begin{aligned}&(a_j - 2D_j x_j^D) - (b_i + 2S_i x_i^S) - t_{ij} \le 0 \\ &(a_j - D_j x_j^D) - D_j x_j^D - (b_i + S_i x_i^S) - S_i x_i^S - t_{ij} \le 0 \\ &p_j^D - D_j x_j^D \le p_i^S + t_{ij} + S_i x_i^S \\ &\quad MR \le MC\end{aligned} \tag{21}$$

In relation $p_j^D - D_j x_j^D \le p_i^S + t_{ij} + S_i x_i^S$, the terms $D_j x_j^D$ and $S_i x_i^S$ constitute a measure of market power of the monopolist and the monopsonist, respectivey. Figure 2 illustrates this cartel spatial model.

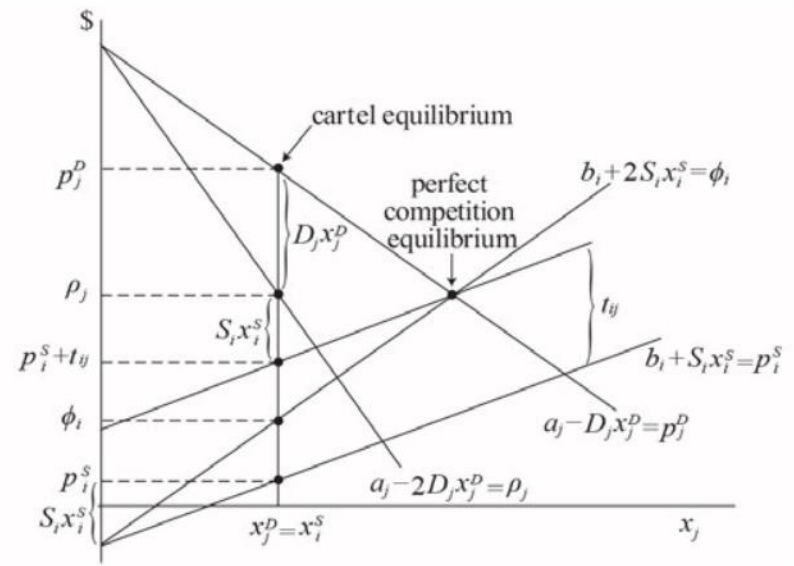


Figure 2. Cartel behavior: monopoly-monopsony

5. Spatial Cournot–Nash Equilibrium: Oligopoly–Perfect Competition

The perfect competition and the cartel (monopoly-monopsony) models represent limiting specifications of spatial equilibrium. In between these two cases there exists a wide series of behavioral performances classified under the two categories of non-cooperative and cooperative imperfect competition rules. We consider here an imperfect competition hypothesis that goes under the name of non-cooperative Cournot-Nash equilibrium. In this context, exporters operate under a perfect competition market. Consumers are price takers, as usual. Each region has one supplier-exporter who makes profit maximizing decisions about output quantities assuming that his choices do not affect the decisions of supplier-exporters in other regions. This is the non-cooperative feature of the model.

In the process toward a general Cournot-Nash model, the i-th region (supplier) primal problem states the maximization of profit, π_i, subject to the supply constraint of the i-th region

$$\begin{aligned}\max \pi_i &= \sum_{j=1}^{R} p_j^D x_{ij} - (b_i + S_i x_i^S / 2)x_i^S - \sum_{j=1}^{R} t_{ij} x_{ij} \\ &= \sum_{j=1}^{R} (a_j - D_j x_j^D)x_{ij} - (b_i + S_i x_i^S / 2)x_i^S - \sum_{j=1}^{R} t_{ij} x_{ij} \\ &= \sum_{j=1}^{R} (a_j - D_j \sum_{k=1}^{R} x_{kj})x_{ij} - (b_i + S_i x_i^S / 2)x_i^S - \sum_{j=1}^{R} t_{ij} x_{ij}\end{aligned} \tag{22}$$

$$D \leq S$$

subject to

$$\sum_{j=1}^{R} x_{ij} \leq x_i^S \tag{23}$$

The non-cooperative hypothesis is expressed by the equation $x_j^D = \sum_k x_{kj}$ which is simply the sum of the supply quantities of all the regions satisfying the demand of the j-th region. The relevant KKT conditions of problem [(22)-(23)] are derived from the Lagrange function

$$L_i = \sum_{j=1}^{R}(a_j - D_j\sum_{k=1}^{R}x_{kj})x_{ij} - (b_i + S_i x_i^S/2)x_i^S - \sum_{j=1}^{R}t_{ij}x_{ij} + \sum_{i=1}^{R}p_i^S(x_i^S - \sum_{j=1}^{R}x_{ij}) \tag{24}$$

and KKT conditions

$$\frac{\partial L_i}{\partial x_i^S} = -b_i - S_i x_i^S + p_i^S \le 0 \tag{25}$$

$$\begin{aligned}\frac{\partial L_i}{\partial x_{ij}} &= (a_j - D_j\sum_{k=1}^{R}x_{kj}) - D_j x_{ij} - p_i^S - t_{ij} \le 0 \\ &= p_j^D - D_j x_{ij} - p_i^S - t_{ij} \le 0\end{aligned} \tag{26}$$

Assuming a positive trade flow on the $i-j$ route, relation (26) becomes an equation (by complementary slackness condition) and $p_j^D = (p_i^S + t_{ij}) + D_j x_{ij} = MC_{ij} + D_j x_{ij}$. In other words, the Nash demand price of the i-th oligopolistic firm in the j-th region is equal to the marginal cost plus the segment (mark-up) $D_j x_{ij}$ (oligopoly power), as indicated in figure 3. The profit of the i-th non-cooperative Nash firm (region) is given by the sum of areas *C* plus *D*.

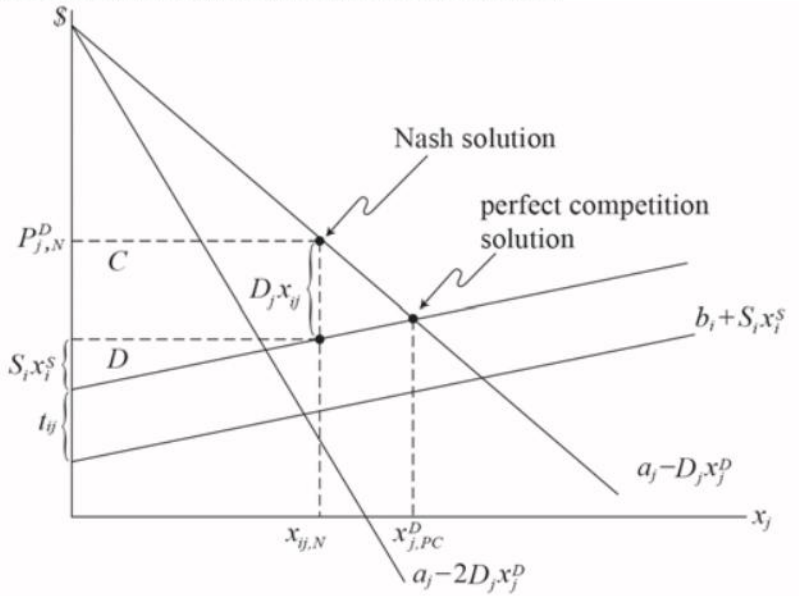


Figure 3. Cournot-Nash (*N*) and perfect competition (*PC*) solutions

From figures 1 and 3 we conclude that the cartel has the lowest cost and the highest demand price together with the lowest supply quantity. Then comes the non-cooperative Nash firm with intermediate cost, quantity and demand price. The perfect competition model exhibits the highest

cost and quantity and the lowest demand price. This implies that the cartel has the highest profit while the Nash firms have lower profit and the perfect competition firms have zero profit. These assertions are valid for the total quantity and profit over all regions while some regions may exhibit Nash prices and profits that are higher than the cartel price and profit and quantities that are lower than the cartel output.

The above discussion pertaining to the non-cooperative behavior of the i-th region (oligopoly firm) guides the specification of the overall spatial Nash equilibrium model that must be expressed as a mathematical programming structure capable to reproduce the necessary conditions (KKT conditions) of each oligopoly firm (region) as stated in relations (25) and (26). Such a model assumes the following specification

$$\max Nash = \sum_{j=1}^{R}(a_j - D_j x_j^D/2)x_j^D - \sum_{i=1}^{R}(b_i + S_i x_i^S/2)x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij}x_{ij} - \sum_{i=1}^{R}\sum_{j=1}^{R} D_j x_{ij}^2/2 \tag{27}$$

subject to

$$D \le S$$

$$x_j^D \le \sum_{i=1}^{R} x_{ij} \tag{28}$$

$$\sum_{j=1}^{R} x_{ij} \le x_i^S \tag{29}$$

with all nonnegative variables. The term $\sum_i \sum_j D_j x_{ij}^2/2$ is required for deriving the correct KKT conditions of each non-cooperative Nash region as demonstrated below. The Lagrange function is stated as

$$\begin{aligned} L = &\sum_{j=1}^{R}(a_j - D_j x_j^D/2)x_j^D - \sum_{i=1}^{R}(b_i + S_i x_i^S/2)x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij}x_{ij} \\ &- \sum_{i=1}^{R}\sum_{j=1}^{R} D_j x_{ij}^2/2 + \sum_{j=1}^{R} p_j^D\left(\sum_{i=1}^{R} x_{ij} - x_j^D\right) + \sum_{i=1}^{R} p_i^S\left(x_i^S - \sum_{j=1}^{R} x_{ij}\right) \end{aligned} \tag{30}$$

with relevant KKT conditions

$$\frac{\partial L}{\partial x_j^D} = a_j - D_j x_j^D - p_j^D \le 0 \tag{31}$$

$$\frac{\partial L}{\partial x_i^S} = -b_i - S_i x_i^S + p_i^S \le 0 \tag{32}$$

$$\frac{\partial L}{\partial x_{ij}} = p_j^D - t_{ij} - D_j x_{ij} - p_i^S \le 0 \tag{33}$$

Relations (32) and (33) are identical to relations (25) and (26) which characterize the Cournot-Nash structure of the spatial problem for the i-th oligopoly firm.

6. Spatial Cournot–Nash Equilibrium: Oligopoly–Oligopsony

The next spatial model deals with non-cooperative Cournot- Nash behavior on both the supply and the export/consumption markets. Building on the reasoning developed in section 6, the profit goal of this behavioral hypothesis takes on the following structure:

$$\max Nash\pi = \sum_{j=1}^{R}(a_j - D_j x_j^D / 2)x_j^D - \sum_{i=1}^{R}(b_i + S_i x_i^S / 2)x_i^S - \sum_{i=1}^{R}\sum_{j=1}^{R} t_{ij} x_{ij} - \sum_{i=1}^{R}\sum_{j=1}^{R} D_j x_{ij}^2 / 2 - \sum_{i=1}^{R}\sum_{j=1}^{R} S_i x_{ij}^2 / 2 \quad (34)$$

subject to

$$D \leq S$$

$$x_j^D \leq \sum_{i=1}^{R} x_{ij} \quad (35)$$

$$\sum_{j=1}^{R} x_{ij} \leq x_i^S \quad (36)$$

As discussed in the previous Cournot-Nash model, the terms $\sum_{i=1}^{R}\sum_{j=1}^{R} D_j x_{ij}^2 / 2$ and $\sum_{i=1}^{R}\sum_{j=1}^{R} S_i x_{ij}^2 / 2$ are required to obtain the correct KKT conditions, which are

$$\frac{\partial L}{\partial x_j^D} = a_j - D_j x_j^D - p_j^D \leq 0 \quad (37)$$

$$\frac{\partial L}{\partial x_i^S} = -b_i - S_i x_i^S + p_i^S \leq 0 \quad (38)$$

$$\frac{\partial L}{\partial x_{ij}} = p_j^D - t_{ij} - D_j x_{ij} - p_i^S - S_i x_{ij} \leq 0 \quad (39)$$

Relation (39), in particular, expresses the behavioral guidelines of this oligopoly-oligopsony hypothesis that is reflected in the fundamental $MR \leq MC$ relation with the following structure $p_j^D - D_j x_{ij} \leq p_i^S + t_{ij} + S_i x_{ij}$. Figure 4 illustrates this Cournot-Nash hypothesis. Compare figure 4 with figure 2.

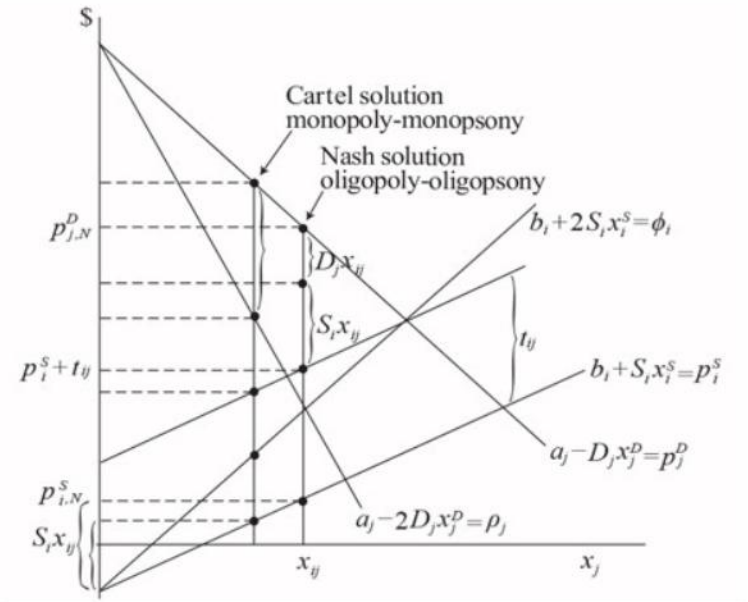


Figure 4. Cournot-Nash equilibrium: oligopoly-oligopsony

7. Numerical Examples of Spatial Equilibria

We present a numerical example of four regions that supply and exchange one commodity through one of the five behavioral hypotheses discussed in previous sections.
These results are exhibited from table 3 to table 8.

We begin with the given information common to the five models. Table 1 presents the intercepts and slopes of the demand and supply functions for the four regions.

Table 1. Demand and supply functions

Regions	Demand Intercept a_j	Demand Slope D_j	Supply Intercept b_i	Supply Slope S_i
A	40.0	1.2	0.4	1.3
B	32.0	0.8	0.2	2.0
U	25.0	0.8	-0.6	1.9
E	38.0	1.1	-0.5	0.6

Table 2 presents the unit transaction costs. Notice that the nominal transaction cost within each region is equal to zero. It could be positive, for example, if we were to consider a commodity priced at farm gate that is sold at a retail store within the same region.

Table 2. Unit transaction costs t_{ij}

Regions	A	B	U	E
A	0.000	2.050	1.620	10.800
B	2.050	0.000	3.240	10.800
U	1.620	3.240	0.000	9.990
E	10.800	10.800	9.990	0.000

The next six tables present the solutions of the five behavioral models discussed in previous sections. For an easy comparison, we group the results of the various optimal quantities and prices according to the order: cartel, non-cooperative Nash, and perfect competition.

Table 3. Trade flow of the five behavioral models

Cartel (monopoly-perfect competition) trade flow, x_{ij}				
	A	B	U	E
A	10.703			
B		8.497		
U		0.757	6.904	
E				13.750
Cartel (monopoly-monopsony) trade flow, x_{ij}				
	A	B	U	E
A	7.699			
B		5.459		
U	0.460	0.767	3.877	
E				11.324
Non-cooperative Nash (oligopoly-pc) trade flow, x_{ij}				
	A	B	U	E
A	6.170	2.597	3.010	
B	1.924	6.353		
U	3.766	2.029	3.455	
E	2.213	1.724	0.112	12.882
Non-coop Nash (oligopoly-oligopsony) trade flow, x_{ij}				
	A	B	U	E
A	4.601	2.434	2.026	0.819
B	2.524	3.464	0.994	0.133
U	3.362	2.196	2.079	0.427
E	2.223	1.766	0.733	10.042
Perfect competition trade flow, x_{ij}				
	A	B	U	E
A	15.398			
B		10.919		
U	0.921	1.535	7.753	
E				22.647

The obvious comment is that the cartel and the perfect competition models exhibit positive trade flows in very few locations. In contrast, the Nash trade flows present positive trade in at least twice as many locations. Further comments concerning each region total demand and supply are presented in connection with tables 4 and 5.

Table 4. Demand of each region, x_j^D

A	B	U	E	Total
Cartel (monopoly-perfect competition) quantity demanded				
10.703	9.254	6.904	13.750	40.610
Cartel (monopoly-monopsony) quantity demanded				
8.159	6.227	3.877	11.324	29.586
Non-cooperative Nash (oligopoly-pc) quantity demanded				
14.072	12.703	6.577	12.882	46.235
Non-coop Nash (oligopoly-oligopsony) quantity demanded				
12.711	9.861	5.832	11.421	39.825
Perfect competition quantity demanded				
16.319	12.453	7.753	22.647	59.173

Observe the cartel, Nash and perfect competition results under the same behavioral hypothesis (that is, monopoly–pc, oligopoly–pc and pc. Or monopoly–monopsony, oligopoly–oligopsony and pc). The cartel chooses the smallest total quantity, the non-cooperative Nash firms choose an intermediate total quantity and the perfect competition model chooses the largest overall quantity, as illustrated in figures 1 and 3. Within each region there may be variations of this trend.

Table 5. Supply of each region, x_i^S

A	B	U	E	Total
Cartel (monopoly-perfect competition) quantity supplied				
10.703	8.497	7.660	13.750	40.610
Cartel (monopoly-monopsony) quantity supplied				
7.699	5.459	5.105	11.324	29.586
Non-cooperative Nash (oligopoly-pc) quantity supplied				
11.777	8.277	9.250	16.931	46.235
Non-cooperative Nash (oligopoly-oligopsony) quantity supplied				
9.880	7.106	8.064	14.775	39.825
Perfect competition quantity supplied				
15.398	10.919	10.209	22.647	59.173

Except for region B, each other region follows the trend where the cartel, the non-cooperative Nash firms and the perfect competitive model exhibit – respectively – an increasing quantity of commodity supplied. Total supply is obviously equal to total demand.

Tables 6 and 7 present the equilibrium prices for the five models.

Table 6. Demand price for each region, p_j^D

A	B	U	E
Cartel (monopoly-perfect compet.) demand prices			
27.157	24.957	19.477	22.875
Cartel (monopoly-monopsony) demand prices			
30.209	27.019	21.899	25.544
Non-coop (oligopoly-pc) Nash demand prices			
23.114	21.838	19.738	23.829
Non-coop (oligopoly-oligopsony) Nash dem. prices			
24.747	24.111	20.335	25.437
Perfect competition demand prices			
20.417	22.037	18.797	13.088

The non-cooperative Nash firms of regions U and E exhibit demand prices that are higher than the cartel prices while, in region B, the non-cooperative Nash demand price is lower than the competitive firms' price.

Table 7. Supply price for each region, p_i^S

A	B	U	E
Cartel (monopoly-perfect comp.) supply prices			
14.314	17.194	13.954	7.750
Cartel (monopoly-monopsony) supply prices			
10.409	11.119	9.099	6.294
Non-cooperative (oligopoly-pc) Nash supply prices			
15.710	16.755	16.954	9.659
Non-coop (oligopoly-oligopsony) Nash supp prices			
13.245	14.411	14.722	8.365
Perfect competition supply prices			
20.417	22.037	18.797	13.088

Supply prices follow – in general (except for region B) – the inverse relation of the demand prices: cartel supply prices are the lowest ones followed by the Nash prices and finishing with the perfect competition prices.

Table 8 presents the profit for each region.

Table 8. Profit of each region

A	B	U	E	Total
Cartel (monopoly-perfect competition) profit				
211.914	140.704	93.871	264.687	711.177
Cartel (monopoly-monopsony) profit				
156.948	90.626	61.530	217.978	527.082
Non-cooperative (oligopoly-pc) Nash profit				
148.474	105.250	111.139	276.812	641.676
Non-cooperative (oligopoly-oligopsony) Nash profit				
81.740	52.107	53.745	194.096	381.688
Perfect competition profit				
0.000	0.000	0.000	0.000	0.000

Consider total profit. The cartel acquires the highest level of total profit followed by the non-cooperative Nash firms and the perfectly competitive firms whose profit is equal to zero by construction. Within the various regions, however, the profit trend is not unique. The pattern of the production and demand quantities and of the corresponding prices depends on the structure of the regional demand and supply functions coupled with the matrix of transaction costs.

Quantities, prices and profit follow a complex regional pattern among the three behavioral assumptions. Total quantities, however, follow the expected trend with cartel presenting on the market the smallest quantity that maximizes its total profit. Then come the quantity and price levels of the non-cooperative Nash firms followed by the quantity and profit levels of the perfectly competitive firms.